# Microbiology Midterm Reviewer

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## Executive Summary

A focused reviewer covering cell structure, bacterial classification, and metabolism for the upcoming examination.

**KEY TAKEAWAYS**

- Gram staining separates bacteria by cell wall
- Aerobic respiration yields more ATP than fermentation
- Organelles compartmentalize eukaryotic metabolism

## Topics

TOPIC 1

### Cell Wall & Membrane

Structural layers that define prokaryotic and eukaryotic boundaries.

#### Peptidoglycan Layer

- Thick in Gram-positive, thin in Gram-negative
- Cross-linked by peptide bridges

TOPIC 2

### Metabolic Pathways

Energy-generating routes central to microbial growth.

#### Glycolysis

- Converts glucose to pyruvate
- Yields 2 ATP per molecule

## Terms & Definitions

| Term | Definition |
| --- | --- |
| **Aerobe** | An organism that requires oxygen for growth. |
| **Fermentation** | Anaerobic ATP generation without an electron transport chain. |
| **Plasmid** | Small circular extrachromosomal DNA. |

## Key Facts & Formulas

| Formula / Fact | Context |
| --- | --- |
| **ATP = $C_{10}H_{16}N_5O_{13}P_3$** | Adenosine triphosphate |
| **$\Delta G°' = -RT \ln K_{eq}$** | Standard free-energy change |